

Digital Data-Capture Ecosystem for Traceable OEE Analysis in a Flexible Manufacturing Cell

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Abstract—This paper presents a digital data-capture and integration ecosystem for standardized, traceable Overall Equipment Effectiveness (OEE) analysis in a flexible manufacturing cell. The study was motivated by the limitations of the previous manual workflow, which relied on paper-based records, repeated transcription, spreadsheet-dependent consolidation, and delayed reconciliation across production and failure data. To address this problem, a digital ecosystem based on Microsoft Lists, Power Apps, SharePoint, and Power Query was implemented to enable direct data capture, centralized storage, automated transformation, and integrated analysis. The ecosystem was evaluated through a global inconsistency comparison, a two-proportions test, a paired comparison between original and corrected OEE values, and deterministic sensitivity analyses. The results showed that the global inconsistency rate decreased from 43.43% in the 2025 manual system to 2.25% in the 2026 digital system. In addition, the pipeline improved repeatability, traceability, and efficiency of the OEE calculation process. The main contribution of this work is to demonstrate that reliable OEE analysis in a flexible manufacturing cell depends not only on the KPI formula, but on standardized data capture, clear programmed-time logic, and an integrated digital workflow that improves traceability, repeatability, and analytical reliability.

Keywords—Overall Equipment Effectiveness OEE, data capture standardization, manufacturing data traceability.

I. INTRODUCTION

Overall Equipment Effectiveness (OEE) is increasingly used in Industry 4.0 environments as an input for forecasting, optimization, and production decision support. While advanced forecasting models require reliable operational data [1], and process digitization in SMEs integrates OEE with value stream mapping [2], partial digitization often leaves data vulnerable to manual transcription errors. Furthermore, process mining and bottleneck analysis rely heavily on structured data from enterprise Manufacturing Execution Systems (MES) like SAP ME [3], which are often financially and technically unfeasible for SMEs and educational laboratories. Consequently, the usefulness of OEE depends not only on its mathematical structure, but on the quality and traceability of the underlying records.

This issue is reinforced by data-quality research identifying accuracy, completeness, consistency, and timeliness as critical dimensions to avoid "garbage in, garbage out" scenarios [4], and by industrial cyber-physical systems (ICPS) literature emphasizing that data integrity and traceability are central requirements for reliable information systems [5]. Together, these studies support the need to treat OEE calculation as a data-governance problem, not only as a performance-measurement problem.

To collect operational data in resource-constrained environments, research has explored low-code/no-code platforms, demonstrating that students can develop digital manufacturing systems to support smart-manufacturing learning [6]. However, while such work focuses primarily on pedagogical outcomes and system architecture, the present study focuses on measurement validity: evaluating whether a digital data-capture ecosystem can reduce inconsistency and support reliable OEE analysis.

This paper hypothesizes that a standardized digital ecosystem can reduce source-record inconsistency and improve traceability in a flexible manufacturing cell. This emerged from limitations in the previous manual workflow, where data was captured on paper, transcribed into spreadsheets, and corrected after delayed reconciliation. In this study, the term *workflow* is defined as the end-to-end data pipeline required to obtain the final OEE value, encapsulating source capture, transcription, consolidation, and KPI calculation. As new cycles were analyzed, this manual workflow became unsustainable due to repeated reconstruction efforts and untraceable error origins.

The present study addresses an upstream problem that precedes advanced OEE analytics: the integrity and traceability of the operational records from which the indicator is calculated. Accordingly, this work evaluates the proposed digital ecosystem as a measurement-data architecture rather than solely as a software implementation. Its scientific contributions are threefold: (1) the operationalization and quantitative comparison of source-record inconsistency before and after digital standardization; (2) the empirical evaluation of whether corrections to manually captured data materially alter the OEE values at the corresponding analytical level; and (3) the deterministic characterization of how programmed-time and failure-

This work used non-identifiable operational production, failure, and quality records. Funding by FlexLab Research Project.

duration recording errors propagate through the performance and availability components of OEE.

Unlike previous studies that focus on OEE forecasting [1], partial SME digitization [2], heavy MES process mining [3], or low-code pedagogical outcomes [6], this work empirically validates the integration of accessible commercial tools as a traceable, agile data-governance pipeline. By doing so, it bridges the gap between expensive enterprise MES and unreliable manual methods, ensuring data veracity in educational and SME flexible manufacturing cells.

II. METHODS

A. Study context

The study was conducted in an educational flexible manufacturing cell used for production and academic training activities. OEE was monitored through its three standard components: Availability, Performance, and Quality. The calculation model for the study is defined in section II-F.

In the original workflow, production, failures, and quality observations were recorded separately by shift and later consolidated for analysis. Over time, this process evolved from a paper-based and spreadsheet-dependent system into a digital ecosystem based on Microsoft Lists, Power Apps, SharePoint, and Power Query.

B. Problem diagnosis

In 2025 the first version of the OEE capture process consisted of the following stages:

1. Raw operational data was captured by operators on paper forms.
2. Handwritten data was later transcribed manually into spreadsheet files by team members.
3. Additional calculated columns were added to obtain derived indicators.
4. The resulting document was analyzed manually and necessary corrections were made.

After the first round of analysis, it was discovered that the time and effort required to process the year's data could not be replicated easily due to the high consolidation time and heavy formula management.

C. Iterative Development

The digital ecosystem was developed iteratively in response to successive bottlenecks identified in the original OEE measurement process. The first stage consisted of paper-based records, in which production, failure, and quality data were captured manually by operators and later transcribed into spreadsheets. This approach enabled initial OEE calculation, but it generated duplicated effort, delayed error detection, and weak traceability because the origin of inconsistencies could not be clearly attributed to either the source record or the transcription process.

Microsoft Lists was then introduced to structure the production, failure, and quality datasets in a centralized digital repository. This improved data organization and provided a more consistent storage format; however, if operators continued recording information first on paper and later transferring it to the digital list, the double-entry problem

remained. To address this limitation, Power Apps was implemented as the direct capture interface, allowing operators to enter production, failure, and quality information directly into the structured digital records.

Once direct capture was established, the remaining bottleneck was the manual consolidation and transformation of the data for OEE analysis. Power Query was therefore incorporated as the integration layer to retrieve the updated records, clean and transform the datasets, merge production, and failure information by analytical unit, and generate the calculated fields required for the OEE structure. In this way, the ecosystem evolved from isolated manual records into a connected data pipeline in which each tool addressed a specific limitation exposed by the previous stage.

D. Digital Ecosystem Architecture and Data Flow

The digital ecosystem was structured as a four-layer architecture designed to support data-quality dimensions relevant to the OEE measurement process—particularly consistency, completeness, validity, and timeliness [4]—throughout the data pipeline. The physical flexible manufacturing cell generates operational events of interest, including production, failure, and quality records. These events are captured, stored, transformed, integrated, and finally converted into the variables required for OEE calculation, creating a continuous digital path from the operational source record to the resulting indicator.

- I. Capture Layer (Power Apps): direct operator interface to eliminate the paper-to-spreadsheet transcription step and standardize data entry at the source. Rather than relying primarily on free-text fields, the application uses constrained controls such as predefined selections for product identification, shift assignment, and failure classification. Temporal information is captured through time-selection controls and converted programmatically into DateTime values before submission.
- II. Storage Layer (Microsoft Lists / Sharepoint): The validated records are written directly to Microsoft Lists through Power Platform connectors. Four structured lists constitute the centralized source records of the ecosystem: *OEE_PROD* for production records by shift and product, *OEE_FALLOS* for failure events and their temporal context and *OEE_CALIDAD* for summarized quality information. Defined field types and common analytical identifiers—particularly date, shift, and product—provide a consistent structure for subsequent reconciliation and improve traceability between the operational event and the analytical record [5].
- III. Transformation and Integration Layer (Power Query): It retrieves the SharePoint lists through dedicated connectors, standardizes data types, and links production, failure, and quality records using their common analytical identifiers functioning as the automated Extract, Transform, and Load (ETL) layer. Traceability is also preserved during correction. When an inconsistency is identified during transformation or review, the corresponding source record can be located

in Microsoft Lists and corrected at its origin. A subsequent Power Query refresh propagates the corrected information through the consolidated tables and OEE calculation.

IV. Calculation Layer (Excel Data Model): The standardized and consolidated data are loaded into the Excel analytical model, where Availability, Performance, Quality, and final OEE are calculated using the deterministic equations defined in Section II-F. Because the calculations are generated from the transformed source tables, changes made to a traceable source record propagate through the analytical model after refreshing, reducing the need for repeated manual formula manipulation.

Power BI is considered as a subsequent visualization extension of the validated data pipeline.

TABLE I. DIGITAL ECOSYSTEM ARCHITECTURE AND DATA FLOW

Layer (Tool)	Technical Function	Data-Quality / Traceability Mechanism
Capture (Power Apps)	Direct operator input submitted to SharePoint	Constrained selections; DateTime validation; standardized Choice/Person fields
Storage (MS Lists / SharePoint)	Centralized source records: production, failure, quality	Defined schemas, field types, shared date-shift-product identifiers
Transformation (Power Query)	Retrieval, cleaning, restructuring, merging into analytical tables	Type enforcement, null handling, downtime consolidation, traceability
Calculation (Excel Data Model)	Availability, Performance, Quality, and OEE calculation	Deterministic equations; refresh-driven recalculation

E. Operational definition of inconsistency

A source record was classified as inconsistent when it presented at least one verifiable problem affecting coherence across the production and failure datasets. Following the data-quality taxonomy proposed by Ridzuan and Zainon [4], these inconsistencies can be mapped to specific quality dimensions:

- *Completeness violations: missing production duration and failure records without associated production context.*
- *Consistency violations: missing product identification in failure logs and mismatch between production and failure product identifiers.*
- *Validity violation: programmed-time values inconsistent with the planned-pallet rule and records requiring manual remapping to reconcile sources.*

This categorization enables a structured diagnosis of where data quality degrades in the measurement pipeline and provides a formal basis for comparing the manual and digital workflows beyond a simple error count.

F. Calculation Model for OEE and Validation metrics

Overall Equipment Effectiveness was used as the main performance indicator to evaluate the flexible manufacturing cell and was calculated as:

$$OEE = A \times P \times Q \quad (1)$$

Where (*A*) is availability, (*P*) is performance, and (*Q*) is quality.

Availability was calculated using the available production time and the consolidated downtime associated with failure records:

$$A = \frac{T_{available} - T_{down}}{T_{available}} \quad (2)$$

Where $T_{available}$ is the available production time calculated by the formula in which the factor 24×60 converts date-time differences stored in fractions of a day into minutes and where “*j*” is the corresponding production record:

$$T_{available} = (\text{production}_{end_j} - \text{production}_{start_j}) \times 24 \times 60 \quad (3)$$

And T_{down} is equal to the sum of the consolidated failure-duration records, in minutes:

$$T_{down} = \sum_{k=1}^{n_c} T_{consolidated_k} \quad (4)$$

Where n_c is the number of consolidated downtime intervals. And the duration of each consolidated interval “*k*” if two or more failure events overlapped is:

$$T_{consolidated,k} = (t_{end_k} - t_{start_k}) \times 24 \times 60 \quad (5)$$

The Time to Repair each failure event “*j*” was calculated as the elapsed time between failure identification and resolution:

$$TTR_j = (t_{resolved_j} - t_{identified_j}) \times 24 \times 60 \quad (6)$$

Performance was calculated as the ratio between executed pallets and planned pallets:

$$P = \frac{N_{executed}}{N_{planned}} \quad (7)$$

Where $N_{executed}$ is the number of pallets completed and $N_{planned}$ is the number of pallets expected according to the programmed production time:

$$N_{planned} = \frac{60 \times T_{programmed}}{t_s} \quad (8)$$

In which t_s corresponds to the standard production time in minutes/pallet.

Planned pallets were obtained from the programmed-time rule established for the cell. This relation is important because recording errors in programmed or available time modifies the number of planned pallets and therefore affects the performance component of OEE.

Quality was calculated from the inspected production quantity and the discarded quantity:

$$Q = \frac{N_{sample} - N_{discarded}}{N_{total}} \quad (9)$$

To evaluate data integrity, the global inconsistency rate was calculated for each workflow: using the following formula:

$$IR_y = \frac{E_y}{N_y} \times 100 \quad (10)$$

Where IR_y is the inconsistency rate for year y . E_y is the number of inconsistent records and N_y is the total audited records.

To evaluate whether data correction affected the OEE indicator, paired differences were calculated between the original and corrected 2025 values:

$$d_i = OEE_{1,0i} - OEE_{2,0i} \quad (11)$$

Where d_i is the paired difference for observation i .

Finally, the sensitivity analysis was expressed using normalized OEE values. For each error scenario, the recalculated OEE was compared with the baseline value:

$$Relative\ OEE = \frac{OEE'}{OEE_0} \times 100 \quad (12)$$

Where OEE_0 is the baseline and OEE' is the recalculated value under an error scenario.

G. Analytical procedure

Table II summarizes the analyses used in the study and their purpose. The two-proportions test was applied to the inconsistency rates defined in (10). The paired comparison between original and corrected OEE values was based on the difference defined in the paired (11). The deterministic sensitivity analyses used the OEE structure in (1) – (9) and the normalized indicator in (12) to evaluate how recording errors propagate through performance and availability.

TABLE II. VALIDATION METHOD OF DIGITAL OEE ECOSYSTEM

Analysis	Purpose
Global inconsistency rate	Quantify source-record inconsistency before and after digitalization
Two-proportions test	Compare inconsistency rates between 2025 and 2026
Anderson-Darling test	Assess normality of paired OEE differences to determine appropriate comparison method
Paired t-test	Compare original vs. corrected 2025 OEE values
Sensitivity analysis: programmed time	Measure the effect of programmed-time error on performance and OEE
Sensitivity analysis: Downtime error	Measure the effect of downtime error on availability and OEE
Data-governance comparison	Compare manual and digital workflows beyond the KPI

III. RESULTS

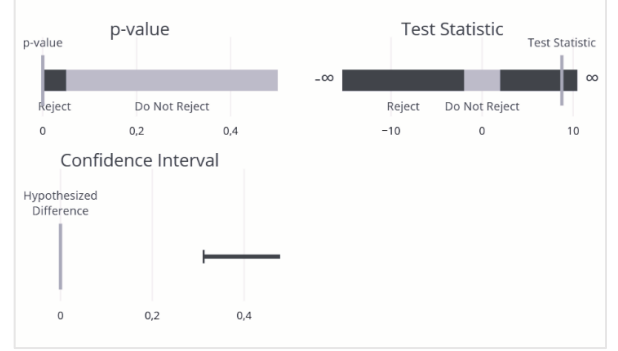
First, the study quantifies the reduction in source-record inconsistency between the manual 2025 workflow and the digital 2026 ecosystem. Second, it evaluates whether the observed data-quality problems affected the OEE values calculated from the 2025 dataset. Finally, deterministic sensitivity analyses are used to explain how recording errors propagate through the OEE structure, specifically through performance in the case of programmed-time error and through availability in the case of Downtime error.

A. Reduction in source-record inconsistency

TABLE III. GLOBAL INCONSISTENCY RATE

Data	System	
	2025 (manual)	2026 (digital)
Production log errors	12	4
Failure log errors	31	0
Total Errors	43	4
Total Records	99	178
Overall Inconsistency Rate	43.43%	2.25%

FIGURE I. TWO-PROPORTIONS TEST OF SOURCE-RECORD INCONSISTENCY, 2025 VS. 2026

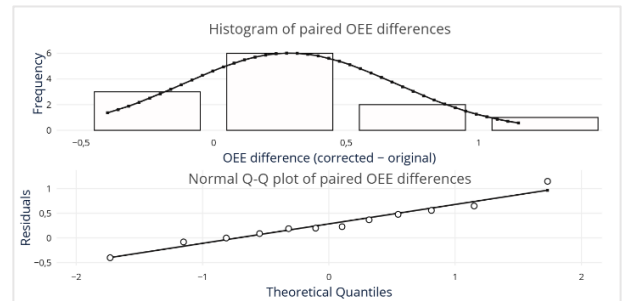


The manual 2025 system exhibited a substantially higher inconsistency rate than the digital 2026 system. The difference was statistically significant according to the two-proportions test, indicating a significantly lower proportion of inconsistent source records under the 2026 digital workflow. Therefore, the proposed workflow not only replaced manual recording, but also improved the integrity of the analytical input used for OEE calculation.

B. Effect of correction on 2025 OEE values

Before comparing the original and corrected 2025 OEE values, the distribution of the paired differences was evaluated to determine the appropriate paired comparison method.

FIGURE II. NORMALITY ASSESSMENT OF PAIRED OEE DIFFERENCES



The Anderson-Darling result did not reject normality of the paired differences, indicating that the comparison between original and corrected OEE values could be assessed using a parametric paired test.

TABLE IV. DESCRIPTIVE STATISTICS PAIRED T-TEST

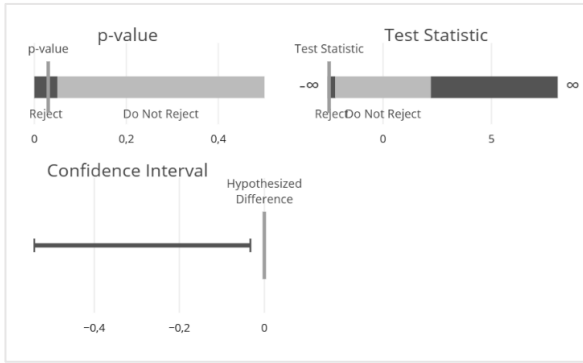
Sample	N	Mean	StDev	SE Mean
OEE 2025 1.0	12	0.4358	0.2728	0.0788
OEE 2025 2.0	12	0.7225	0.3394	0.0980

The paired comparison showed a statistically significant difference between the original and corrected OEE values.

TABLE V. ESTIMATION FOR PAIRED DIFFERENCE

Mean	StDev	SE Mean	95% CI for population mean of (OEE 2025 1.0 - OEE 2025 2.0)	T-Value	P-Value
-0.287	0.400	0.115	(-0.541, -0.033)	-2.48	0.03

FIGURE III. PAIRED T-TEST OF ORIGINAL AND CORRECTED 2025 OEE



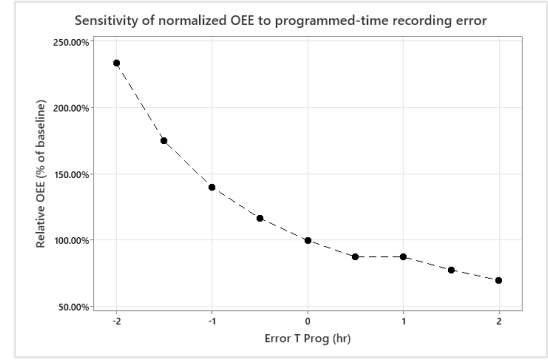
This indicates that the problems present in the manual 2025 workflow were not limited to documentation burden or weak traceability; they directly affected the numerical value of the performance indicator. More importantly, the effect is relevant at the operational decision level because shift- and product-level OEE values can be distorted even when an aggregated indicator appears acceptable. Therefore, preserving traceability by date, shift, and product is necessary to prevent record-level inconsistencies from being concealed by summarized performance views.

The statistical significance of the paired t-test demonstrates that the inconsistencies present in the manual workflow directly distorted the numerical value of the performance indicator. This highlights a fundamental shift in error management between the two ecosystems. In the 2025 manual system, error control at the source was limited and inconsistencies were generally identified during subsequent reconciliation. In contrast, the 2026 digital ecosystem shifts the paradigm from delayed correction to proactive validation and traceable correction. At the capture layer, this is achieved through constrained interface controls and programmatic temporal logic that prevents impossible time entries. At the transformation layer, the ETL engine automatically consolidates overlapping failure intervals and restructures wide-format quality data, ensuring that the remaining inconsistencies are operational edge cases rather than transcription errors.

C. Sensitivity of OEE to programmed-time recording error

Because programmed time determines the number of planned pallets, any error in this input propagates directly through the performance component of OEE. A deterministic sensitivity analysis was therefore conducted to quantify how deviations in programmed time recording affect relative OEE.

FIGURE IV. SENSITIVITY OF RELATIVE OEE TO PROGRAMMED-TIME RECORDING ERROR

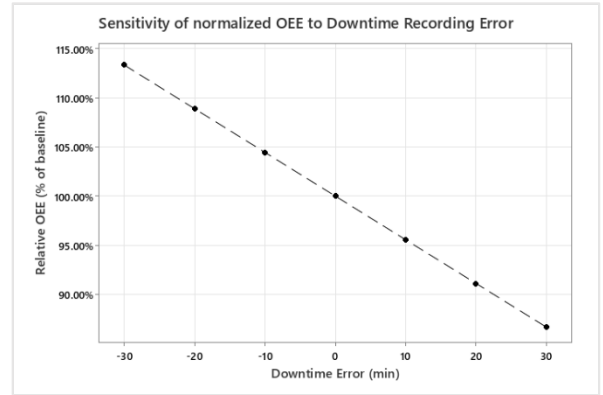


The sensitivity profile shows that underestimation of programmed time artificially inflates OEE by reducing the number of planned pallets and increasing the calculated performance ratio. Conversely, overestimation of programmed time reduces relative OEE. This confirms that programmed-time inconsistency does not primarily distort availability but rather propagates through the performance pathway of the indicator.

D. Sensitivity of OEE to Downtime Recording Error

A second deterministic sensitivity analysis was used to evaluate how Downtime recording error propagates through the availability component and modifies final OEE.

FIGURE V. SENSITIVITY OF RELATIVE OEE TO DOWNTIME RECORDING ERROR



Unlike programmed-time error, Downtime error affects OEE through availability. The results show a monotonic effect: negative error in Downtime increases relative OEE, while positive error decreases it. Although the magnitude of the effect is smaller than that observed for programmed-time error, the analysis confirms that downtime-recording inaccuracies remain analytically relevant.

E. Data governance improvement

In addition to the statistical reduction in inconsistency, the digital ecosystem changed the governance of the analytical workflow. The comparison below summarizes the operational differences between the 2025 manual process and the 2026 digital process.

TABLE VI. DATA GOVERNANCE COMPARISON TABLE

<i>Dimension</i>	<i>2025 manual</i>	<i>2026 digital</i>
Source capture	Paper	Direct digital capture
Transcription	Paper → Excel	None
Error traceability	Limited	Source-record traceability
Correction	Paper/Excel reconciliation	Correction at source in Lists
Analytical consolidation	Manual/semi-manual	Refresh-driven Power Query pipeline
Repeatability	Process reconstruction required	Reusable analytical workflow

The digital workflow transformed the analysis process from a manually repeated reconstruction effort into a reusable analytical workflow.

IV. DISCUSSION AND CONCLUSIONS

The reduction in source-record inconsistency from 43.43% to 2.25%, confirmed by the two-proportions test, together with the statistically significant paired difference between original and corrected 2025 OEE values, demonstrates that the data-quality problems of the manual workflow affected not only traceability and administrative effort, but also the numerical value of the indicator itself. In practical terms, decisions based on the original manual dataset would have relied on a distorted analytical representation of system performance. By preserving traceability from each operational record to the final indicator, the proposed digital ecosystem decreases the influence of data-processing variability, making the calculated OEE a more dependable representation of the manufacturing process.

The sensitivity analyses provide a mechanistic explanation for this distortion. Programmed time recording error propagates through the performance component by modifying the number of planned pallets, whereas downtime recording error affects OEE through availability. This distinction is important because it shows that capture errors do not influence OEE uniformly. For the present study, the programmed-time pathway is particularly relevant since it reflects the need to standardize production-planning logic together with data capture and integration.

Klimecka-Tatar and Ingaldi [2] demonstrated the value of digitized OEE information for process visibility, while Rudnitskaia et al. [3] used data already available in an MES for process and bottleneck analysis. In contrast, the present study evaluates the preceding measurement problem: how operational records are captured, reconciled, and transformed before OEE becomes available for downstream analysis. The closest architectural comparison is the low-code learning-factory system of Bradley et al. [6]; however, whereas that study evaluates system-development and learning outcomes, the present work quantitatively evaluates source-record consistency and the propagation of recording errors into OEE. Therefore, the contribution lies not in the novelty of the individual software tools, but in their validated integration as a traceable measurement-data pipeline.

The study has several limitations that contextualize its findings. First, the evaluation was conducted in a single educational flexible manufacturing cell, a scope dictated by

the resource and budget constraints typical of academic pilot projects. Second, the digital ecosystem was evaluated against the previous manual workflow rather than an independent reference measurement system. Third, the paired OEE comparison was limited to valid observations where both original and corrected values were available, and the sensitivity analyses were deterministic. Finally, while the low-code Microsoft technologies used here are suitable for rapid deployment in resource-constrained environments, they may present scalability limitations in high-frequency industrial settings. Nevertheless, the underlying data-governance architecture and ETL integration workflows are conceptually transferable to enterprise databases or traditional Manufacturing Execution Systems (MES).

From the perspective of intelligent systems, robotics, and automation, the main contribution of this work is the integration of a physical flexible manufacturing cell with a traceable digital information layer capable of supporting more reliable operational analytics. The paper does not claim that digitalization alone improves production performance. Instead, it demonstrates that digital standardization supports more traceable and analytically reliable OEE measurement, providing a stronger basis for identifying process-improvement opportunities that may otherwise remain obscured by manual transcription, delayed reconciliation, and repeated spreadsheet rework. Future work should validate the ecosystem over additional production cycles, compare automated OEE calculations with a validated reference calculation through an equivalence test, and extend the workflow to other manufacturing cells or processes once continuous operational stability is achieved.

V. AUTHOR'S CONTRIBUTION

I.D.: Study design, data analysis, manuscript drafting.
F.B.M. & K.H.C.: Conceptualization, manuscript review.
P.R.V.: Manuscript review, final editing.

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